Utsav Paneru

AI/ML engineer specializing in LLM fine-tuning, with hands-on experience applying QLoRA, instruction tuning, and the Hugging Face stack (Transformers, PEFT) to real fine-tuning projects on Mistral-7B and BERT. Published research on text-style transfer using parameter-efficient fine-tuning across encoder-decoder and decoder-only architectures. Native Nepali speaker with direct interest in extending LLM capability to Devanagari and low-resource language settings. Comfortable building and evaluating models end-to-end in Python, from data curation through deployment.

CONTACT

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Kalanki , Kathmandu

EDUCATION

Bachelors in Computer Engineering
Kathmandu Engineering College / Kathmandu / February, 2024 - Present

SKILLS

- LLM Fine-Tuning: LoRA/QLoRA, instruction tuning (SFT), 4-bit quantization, Hugging Face Transformers
- Languages & Tools: Python, PyTorch, FastAPI, Next.js, REST APIs, JWT Auth

EXPERIENCE

AI Engineer Intern

MetaMinds / Lalitpur / November, 2025 - March, 2026

- Built a chatbot using NLP-based entity extraction to parse user queries and route structured data, working directly with the Hugging Face Transformers stack for model training and inference. Iterated on data preprocessing and evaluation to improve model accuracy on noisy, domain-specific accounting text.
- Fine-tuned a BERT-based model for accounting document analysis, adapting a pretrained transformer to domain-specific text classification and structured information extraction.
- Iterated on data preprocessing and evaluation to improve model accuracy on noisy, domain-specific accounting text.

Web-Development and API integration

Vizon-x (project based / Lalitpur / May, 2025 - September, 2025

- Integrated an LLM API and a RAG (Retrieval-Augmented Generation) pipeline into a production web application, connecting a vector retrieval layer to an LLM for context-grounded responses.
- Built supporting backend and frontend components to surface AI-generated outputs within the existing application, working across the full web stack.

PROJECTS

settleAI — Nepali Voice Assistant

- Engineered a full-duplex Nepali voice AI pipeline combining STT, RAG, LLM streaming, and TTS.
- Built VAD, WebSocket streaming, barge-in, semantic caching, and GPU inference for real-time interaction.
- Optimized inference to achieve ~16× faster TTS while reducing response latency and unnecessary model calls.
- Designed the system for low-cost production inference (~$0.004 per uncached voice turn) using caching and workload optimization.
- Deployed a distributed AI stack across FastAPI, RunPod GPU workers, ChromaDB, Groq, and Vercel.

github.com/cive202/settleAI-NepaliVoiceAssistance

Neural Style Transfer for AI-Generated Text — Research Paper

- Built an end-to-end pipeline comparing encoder–decoder models (BART, T5) against decoder-only architectures (Mistral, Gemma-2B) for AI-to-human text style transfer.
- Applied QLoRA fine-tuning on Mistral-7B with 4-bit quantization for efficient, scalable training on consumer-grade hardware.
- Designed a custom evaluation framework using 11 linguistic features with a directional shift metric, demonstrating improved human-likeness while preserving semantic fidelity (BERTScore ≈ 0.92).
- Paper: arxiv.org/abs/2604.11687

Models: huggingface.co/cive202 (BART-base, BART-large, Mistral-7B-LoRA)

RAG-Based AI Teaching Assistant

Built a Retrieval-Augmented Generation (RAG) application using Python, Flask, and Cohere embeddings to answer questions from lecture transcripts through semantic search. Deployed the application using Docker and Render.

https://github.com/cive202/ragbased_ai